

YASANTHA NIROSHANA

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Research Interests

Embedded Machine Learning (TinyML), Neuromorphic Computing, Hardware-Software Co-design, and Resource-constrained Neural Network Optimization.

Education

University of Moratuwa

B.Sc. (Hons) in Engineering, Computer Science and Engineering

Aug 2022 – 2026

Moratuwa, Sri Lanka

- CGPA: **3.73/4.00**
- Specialization: **Integrated Computer Engineering**
- Minor: **Pattern Recognition**
- **Selected Coursework:** Pattern Recognition, Machine Vision, Deep Learning for Vision, Introduction to Engineering Optimization, Convex Engineering Design, Machine Intelligence, Image Processing, Embedded Systems and Applications, Robotics and Automation, Embedded Software Engineering, IoT Devices and Applications

Mahinda College

GCE Advanced Level (Physical Science Stream)

2007 – 2020

Galle, Sri Lanka

- Z-Score: **2.562** | Island Rank: **115** | Result: **3A's**

Publications

Niroshana, Yasantha, Weijith Wimalasiri, and Chathuranga Hettiarachchi. “Patient-Aware Contrastive Learning Preserves Per-Patient Structure in RR-Interval Representations”. In: *GlobalSouthML: Bridging Gaps for Underrepresented Researchers in Machine Learning, ICML 2026 Workshop*. arXiv:2606.23570. May 2026. arXiv: [2606.23570](https://arxiv.org/abs/2606.23570). URL: <https://arxiv.org/abs/2606.23570>.

Niroshana, Y., S. Sooriyaarachchi, and I. Gunarathne. *Bee Acoustic Dataset with Environmental Parameters*. Accessed: Apr. 26, 2026. Apr. 2026. DOI: [10.5281/zenodo.19739013](https://doi.org/10.5281/zenodo.19739013). URL: <https://doi.org/10.5281/zenodo.19739013>.

Niroshana, H., W. Wimalasiri, and C. Hettiarachchi. “Peak detection of PPG signals using fixed-point digital filters implemented in VHDL”. In: *Proceedings of the ERU Symposium 2025, Engineering Research Unit, University of Moratuwa*. Accessed: May 01, 2026. Jan. 2025, pp. 80–81. DOI: [10.31705/ERU.2025.38](https://doi.org/10.31705/ERU.2025.38). URL: <https://dl.lib.uom.lk/items/73c999d1-3082-4812-94ae-30e70bf6d68a>.

Kodisinghe, R., Y. Niroshana, and W. Wimalasiri. “A Smart mask for wireless, real-time monitoring of CO2 and humidity”. In: *Proceedings of the ERU Symposium 2025, Engineering Research Unit, University of Moratuwa*. Accessed: Apr. 26, 2026. Jan. 2025. DOI: [10.31705/ERU.2025.6](https://doi.org/10.31705/ERU.2025.6). URL: <https://dl.lib.uom.lk/items/6d7048aa-743f-470d-b7a8-65941aa91edb>.

Research & Technical Projects

NPU-Accelerated Heat Index Prediction | *Embedded ML* | [\[GitHub\]](#)

2025

- Developed an end-to-end pipeline to predict Heat Index using a Luckfox Pico NPU. Designed and trained a model using **Quantization-Aware Training (QAT)**, and deployed the quantized (**INT8**) model.
- Analyzed inference throughput and memory access patterns on the **NPU** while maintaining high regression accuracy.

Predicting AFib at Edge (Ongoing) | *Embedded ML* | [\[GitHub\]](#)

2025

- Advisor: Dr. Chathuranga Hettiarachchi
- Researched lightweight CNN architectures for real-time AFib prediction from PPG sensors. Focusing on model pruning and temporal feature extraction to enable continuous monitoring on wearable devices.

IntelliBee: Acoustic TinyML for Beehive Monitoring <i>IoT, DSP</i> [GitHub]	2024
Designed a resource-efficient acoustic monitoring system to detect swarming via Linear Predictive Coding (LPC) coefficients. Leveraged statistical variance of coefficients as features to replace high-compute FFT/spectrogram methods.	
IoT-Air Quality Monitor <i>Embedded Systems</i> [GitHub]	2024
- Advisor: Dr. Kuttila Gunasekara - Engineered a low-power sensor node integrating TVOC, dust, and particulate matter sensors. Optimized data logging intervals and transmission protocols for long-term outdoor deployment.	
8-Bit Custom NanoProcessor Synthesis <i>Computer Architecture, VHDL</i> [GitHub]	2023
Designed and synthesized a custom 8-bit processor. Implemented an ALU and a custom instruction set capable of being executed on a Basys3 FPGA.	

Experience

University of Moratuwa	2026 – Present
<i>Research Assistant</i>	<i>Moratuwa, Sri Lanka</i>
Research in predictive monitoring — machine learning for physiological signals under tight resource budgets.	
University of Moratuwa	Jul 2025 – Present
<i>Teaching Assistant</i>	<i>Moratuwa, Sri Lanka</i>
- CS3631 - Deep Neural Networks CS4363 - Hardware Description Languages - Previously: CS3283 - Embedded Systems Project CS3340 - Robotics and Automation - Mentored 24+ undergraduate project teams across CS3283/CS3340 — five published at the ERU Symposium 2025; currently co-mentoring 2 ongoing projects.	
Ackcio	Dec 2024 – May 2025
<i>Embedded Software Engineer Intern</i>	<i>Remote</i>
Optimized diagnostic tool for industrial sub-GHz wireless mesh networks. Analyzed network congestion patterns to improve reliability in high-interference geotechnical environments.	
RoboticGen	May 2023 – Present
<i>Lead Software Engineer (Part-time)</i>	<i>Colombo, Sri Lanka</i>
Leading the Edu-Tech software development team, focusing on developing software for a STEM learning.	

Certifications

Embedded Software Essentials Specialization University of Colorado Boulder (via Coursera)	2024
Introduction to Embedded Systems Software Embedded Software and Hardware Architecture	
TinyML Professional Certificate Harvard University (via edX)	2024
Fundamentals of TinyML	
Embedded Machine Learning Edge Impulse	2024
Introduction to Embedded ML Computer Vision with Embedded Machine Learning.	

Awards & Honours

Winner Evolve IoT Research Competition (University of Kelaniya)	Oct 2024
IoT-based smart helmet system using the Thread protocol for the mining industry.	
Second Runner Up Bashaway SLIIT FOSS Community	Oct 2024
National-level Linux scripting competition focused on system administration and open-source workflows.	
Second Runner Up Brainstorm IEEE EMBS, University of Moratuwa	Jul 2024
AI-based physiotherapy assistant that provides real-time feedback and personalized exercise programs.	
Bronze Medal All Island Mathematics Competition (Ministry of Education)	Oct 2019

Invited Talks & Sessions

Introduction to IoT SLIoT 2025, University of Moratuwa	Feb 2025
MoraForesight 1.0 & 2.0: Getting Started with IoT/Robotics	2023 – 2024
Introduction to Arduino Programming Kickstart to Robotics, IEEE Techverse SL	Sep 2022

Leadership & Volunteer Activities

SLIOT Challenge <i>Lead - Selection Committee</i> Led a team in evaluating 200+ industry-scale IoT projects for 700+ contestants, ensuring technical rigor.	Oct 2023 – Apr 2024 <i>University of Moratuwa</i>
MoraXtreme 8.0 <i>Academic Excellence Committee</i> Coordinated algorithmic problem set development for undergraduates preparing for IEEE Xtreme.	Sep 2023 – Oct 2023 <i>University of Moratuwa</i>

References

Prof. Chandana Gamage Senior Lecturer, Dept. of CSE University of Moratuwa chandag@cse.mrt.ac.lk	Dr. Kutila Gunasekera Senior Lecturer, Dept. of CSE University of Moratuwa kutila@cse.mrt.ac.lk
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